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OBJECTIVE

To acquaint myself with the latest trends and developments in the field of Embedded Systems and work towards a challenging career with emphasis on developing my technical skills.

WORK EXPERIENCE

- Currently working In **Cyient Limited. Duration – 5 Months**. (Mar 2023 to till date).
- Worked as a Software Engineer in **Mphasis Limited. Duration – 1.6 years**. (Sep 2021 to Mar 2023s).
- Worked as a Software engineer in **Kernex Microsystems India (Ltd.), Hyderabad. Duration – 2.4 Yrs**. (Jun 2019 to Sep 2021).

PROFESSIONAL SUMMARY

- Graduate with Five years of experience in Embedded Designing, Implementation, Integration and Developments as per client requirements using **Microcontrollers**.
- Good exposure in Embedded Firmware Development using C Programming Language using STM32CUBE IDE for **STM32F767IGK6** ARM Cortex M7 Controller board.
- Good exposure in Embedded Firmware Development using C Programming Language using CCS IDE for **TMS570LS3137** ARM Cortex M7 Controller board.
- Good exposure in Embedded Firmware Development using C Programming Language using Keil IDE for **LPC 4357** ARM Cortex Controller board.
- Good exposure in Embedded Application Development using C Programming Language using Keil IDE for **LPC 2378** ARM 7-TDMI Controller board.
- Strong knowledge of **C, FreeRTOS, LINUX**.
- Well acquainted with all phases of **SDLC.**
- Well conversant with Serial **UART** Protocol, Interrupt Handling, Watchdog Timers, **GPIO** Programming.
- Experience in **CAN, RS232, RS485, I2C, SPI** Protocol for serial data communication.
- Strong in SPI Application driver, ARM controller interfacing with Memory using **SPI** Protocol.
- Strong in I2C Application driver, ARM controller interfacing with **ADC, RTC** using **I2C** Protocol.
- Strong in interfacing of peripherals like **GPS**, **SENSOR**.
- Developed coding standards to ensure compliance with the **MISRA** rules.
- Experience in implementation and interfacing with peripheral, data processing and handling various critical safety application software for Railway industry.
- Good knowledge in kernel concepts like Multithreading, Inter Process Communication

(Pipe, FIFO, Message Queue, Shared Memory), Signal, Mutex and Semaphores, Memory Management etc.

- Good in code debugging Skill and Knowledge of **JTAG** and **U-LINK**.
- Knowledge on Safety integrity level (SIL-4) Certification
- Flexibility to adapt to new technological environment.
- Strong analytic and problem-solving capabilities.
- Involved in Training to Loco Pilots about the functionality and operating of TCAS device and its sub systems like OCIP (Operation cum Indication Panel) and BIU (Brake Interface Unit).

ACADEMIC CREDENTIALS:

- Graduate in Electronics and Telecommunication from St.mark Educational Institutions Society Group of Institutions Affiliated to Jawaharlal Nehru Technological University ,Anantapur in 2018 with 70.57%.
- Intermediate Education from Narayana Jr college,Tirupathi in 2014 with 90.50%.
- SSC from Nagarjuna High School in 2012 with 9.7 GPA.

COMPUTER AND TECHNICAL SKILL

- ❖ **Languages** : C, Embedded C, Embedded C++.
- ❖ **Micro Controllers** :STM32F767,TMS570LS3137,LPC4357,LPC 2378, 8051 core family.
- ❖ **Communication Protocols** : CAN, I2C, UART, SPI, ARINC429.
- ❖ **Networking Protocols** : TCP/IP
- ❖ **Compilers** : GCC.
- ❖ **IDE's** : Keil, Eclipse, visual studio, CCS, STM32CUBE
- ❖ **Static code Analysis tools** : Axivion ,Sonar Qube ,Pc-Lint
- ❖ **Version controls** : Tortoise SVN , Git

PROJECT WORK

1) <u>Implementation of NAVCOM</u>

Client : NAL (National Aerospace Laboratories).

Role : Development and Testing.

IDE : CCS, STM32CUBE.

Hardware : STM32F767IGK6, TMS570LS3137.

Language : C**,** Embedded C

Protocols : CAN**,** I2C and UART, IP, UDP.

Team : 3

Description: Entirely new and vital Aerospace safety system under development with NAL. The requirements for this system are derived from the experience with GARMIN SAFE LANDING systems. NAVCOM is a Safety Landing Project for Commercial Aircrafts.NAVCOM uses the Specific RF Range Frequencies to Communicate with the Ground Staff while Landing and Take-off. And also Provides the HMI (Human Machine Interface) For Pilots for Better Understandings.

<u>Responsibilities:</u>

- Contributed in making the modules According to Design standards (DAL-A, DAL-B).

- Developed the SPI Driver For Connecting the PFD and TRANSPONDER For Data Sending and Receiving.
- Developed the UART Driver for interfacing the GPS Module to get the GPS Coordinates.
- Implemented the TFTP Application for Data Receiving and Sending while Flight in the Ground.
- Brought in all the processes to be followed for the project to run smoothly as a team Member

3) Implementation of StativCtrl & AchsCtrl

Client : Carl-Zeiss (Germany)

Role : Coding

Software : Keil

Hardware : LPC 4357.

Language : C, Embedded C

Protocols : CAN, I2C and UART

Team : 2

Description: The project involves the medical device Kinevo Surgical Microscope whose code needed to be Design Doc Strandards. Worked on the Coding according to Design Doc Standards enablement for the various floor stand modules (AchsCtrl M0, AchsCtrl M4, StativCtrl, Boot loader)

Responsibilities:

- Contributed in making the modules According to Design standards.
- Also made the code according to Zeiss coding guidelines.
- Helped in improving the code from past experiences apart from Misra violations.
  E.g, Function return values checking, function used or not in globally
- Brought in all the processes to be followed for the project to run smoothly as a team Member

4) Implementation of Oberhausen SCA

Client : Carl-Zeiss

Role : Coding.

Software : Keil

Hardware : LPC 4357.

Language : C, Embedded C

Protocols : CAN, I2C and UART

Team : 4

Description: The project involves the medical device Kinevo Surgical Microscope whose code needed to be Misra compliant. Worked on the Misra compliance enablement for the various floor stand modules (AchsCtrl M0, AchsCtrl M4, StativCtrl, Boot loader)

Responsibilities:

- Contributed in making the modules Misra compliant.
- Also made the code according to Zeiss coding guidelines.
- Helped in improving the code from past experiences apart from Misra violations.
  E.g, creating Null pointer checker, replacing macros with void arguments and many more.
- Brought in all the processes to be followed for the project to run smoothly as a team Member

5) Implementation of Loco TCAS System

Client : RDSO (Research Designs and Standards Organization).

Role : Designing and Coding.

Software : Keil

Hardware : LPC 2378.

Language : C, Embedded C

Protocols : CAN, I2C and UART

Team : 4

Certification : Safety Integrity Level 4(SIL 4)

Description: Entirely new and vital railway safety system under development with RDSO. The requirements for this system are derived from the experience with ACD systems. In TCAS a customized TDMA protocol is used to manage communication which offers better reliable communication channel than compared with carrier sense alone. Station TCAS allots slots for Loco TCAS for transmission. TCAS uses RF Tags and relies less on GPS for absolute position and Track identification.

1. Developed **UART** driver for Radio modem testing.
2. Developed **UART** driver for **GPS** testing.
3. Developed **I2C** driver for RTC updating.
4. Developed **I2C** driver for **ADC** and **DAC** interfacing.
5. Developed **SPI** driver for Loco parameters and Speed Parameters for Loco TCAS.
6. Developed Software Modules for PSR, SSP, Gradients, Loopline.

Responsibilities:

- Designed and implemented the specification outlined by RDSO for prototype along with the team.
- Application programming development of Loco TCAS.
- Unit testing of functionalities.
- Code Analysis & Event-log analysis and fixing the bugs
- Preparation of user manual

6) Implementation of RCU (Radio Communication Unit)

Client **:** SCR (South Central Railway).

Role **:** Designing and Coding.

Software **:** Keil.

Hardware **:** LPC 2378

Language **:** Embedded C.

Protocols **:** UART, I2C, SPI.

Certification **:** Under processing Safety Integrity Level 4(SIL 4)

Description: Radio Communication Unit is Communication strength finding and distance measuring between the radio modems project of South-central railways. The main task is to measuring the radio signal strength and distance between the two radio modems by taking the GPS coordinates as inputs. The whole data can be stored into external flash using SPI Protocol. The unit can be further used as to test the communication of newly installed stations in field.

1. Developed **UART** driver for interfacing of Radio Modem for exchanging of data between two Radio Communication Units.
2. Developed UART driver for interfacing the GPS Module to get the GPS Coordinates.
3. Preparing maps using arc map by taking the log data.

Responsibilities:

- Responsible for design and development of software.
- Interfacing the peripherals, functional testing.
- Unit and Function Level testing.
- Maintains code as per MISRA -2012 Standard.
- Design and developed technical documentation of Project.

PROFESSIONAL SKILLS

- Good communication and interpersonal skills along with being flexible and versatile to new environments and ability to work with teams as well as individually.
- Excellent analytical and problem-solving skills.
- Motivation to take independent responsibility as well as ability to contribute and be a productive team member.

TRAININGS

- Post graduate diploma in Advance Embedded systems from **Vector India Pvt Ltd** Hyderabad in Jun 2018 to Jan 2019.

PERSONAL DETAILS

Name	**:** Hari Kumar Reddy.
Date of Birth	**:** 15/06/1997
Nationality	**:** Indian.
Marital Status	**:** Single.
Languages	**:** English, Telugu and Hindi.